

Notes: Milgrom & Watt (RES, 2026)

A Walrasian Mechanism with Markups for Nonconvex Markets

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1 Big picture

- **Question.** How can market designers use anonymous linear prices in markets with nonconvexities, such as fixed costs, indivisibilities, start-up costs, or other lumpy production technologies, when ordinary Walrasian equilibrium may fail to exist?
- **Main contribution.** The paper introduces **markup equilibrium**, an extension of Walrasian equilibrium in which sellers receive producer prices p while buyers pay marked-up prices $(1 + \alpha)p$.
- **Core idea.** The markup creates a budget surplus that can pay for excess production caused by nonconvexities. This avoids “uplift” payments and avoids assigning agents bundles they do not demand at the prices they face.
- **Key conceptual innovation.** The paper develops a **Bound-Form First Welfare Theorem**: for any feasible allocation and any price vector, welfare loss is bounded by rationing losses plus the value of excess production.
- **Why this theorem matters.** It gives an approximate efficiency certificate even when exact Walrasian support is impossible. If rationing is small and wasted supply is small, the allocation is nearly efficient.
- **Markup equilibrium properties.** Markup equilibria are resource-feasible, budget-feasible, individually rational, and envy-free at the buyer and seller prices.
- **Computation.** The simple markup mechanism computes a nearly efficient outcome using convexified optimization problems and a one-dimensional search over the markup α .
- **Large-market result.** Under regularity assumptions, the markup α is $O(1/|N|)$, the deadweight loss is $O(1)$, and the percentage welfare loss is $O(1/|N|)$.
- **Incentive result.** The markup mechanism inherits large-market incentive properties from Walrasian mechanisms. It is approximately interim incentive compatible and, under strong monotonicity, approximately ex post incentive compatible.
- **Economic takeaway.** Linear price mechanisms can survive nonconvexities if buyers pay a small markup over producer prices. The markup replaces subsidies and uplift payments with a budget-balanced rule that remains close to Walrasian pricing.

2 Model primitives and measurement

2.1 Agents, goods, and quasilinearity

- There is a finite set of buyers N and a finite set of firms or sellers F .
- There are L varieties of goods and a numeraire good, money.
- A buyer n chooses a consumption bundle $x_n \in X \subset \mathbb{R}_+^L$.
- Buyer n has quasilinear utility

$$U_n(x_n, t) = u_n(x_n) - t,$$

where t is the payment.

- A seller f chooses a production bundle $y_f \in Y \subset \mathbb{R}_+^L$.
- Seller f has cost function $c_f : Y \rightarrow \mathbb{R}_+$ and profit

$$\pi_f(y_f, t) = t - c_f(y_f).$$

Interpretation: Quasilinearity allows the authors to express welfare as total value minus total costs and to study mechanisms through prices and transfers. It removes income effects and focuses attention on nonconvexities in preferences and production.

2.2 Economy and type distribution

An economy is

$$\mathcal{E} = \langle N, (u_n)_{n \in N}, F, (c_f)_{f \in F} \rangle.$$

The paper also describes economies through normalized empirical type distributions:

$$\mu(u_n) = \frac{\#\{\text{buyers in } \mathcal{E} \text{ with valuation } u_n\}}{|N|},$$

$$\chi(c_f) = \frac{\#\{\text{sellers in } \mathcal{E} \text{ with cost function } c_f\}}{|F|}, \quad \phi = \frac{|F|}{|N|}.$$

Interpretation: This representation is useful for large-market limits. The paper studies sequences of economies where the empirical distribution of types stabilizes while the number of participants grows.

2.3 Feasibility, surplus, and first best

An allocation is

$$x = ((x_n)_{n \in N}, (y_f)_{f \in F}).$$

It is resource-feasible if

$$\sum_{n \in N} x_n \leq \sum_{f \in F} y_f.$$

The surplus of allocation x is

$$S(x) = \sum_{n \in N} u_n(x_n) - \sum_{f \in F} c_f(y_f).$$

The efficient surplus is

$$S^* = \max_{x \in X} S(x),$$

where X is the set of feasible allocations. The deadweight loss of allocation x is

$$S^* - S(x).$$

Interpretation: The full efficiency benchmark is the standard quasilinear surplus maximum. The challenge is to implement or approximate this benchmark with prices when individual feasible sets and cost functions are nonconvex.

2.4 Buyer and seller pricing rules

The paper allows different price vectors for buyers and sellers:

- buyers pay $p^b \cdot x$ for bundle x ;
- sellers receive $p^s \cdot y$ for producing y .

Buyer n 's demand correspondence is

$$D_n(p^b) = \arg \max_{x \in X} \{u_n(x) - p^b \cdot x\},$$

with indirect utility

$$\hat{u}_n(p^b) = \max_{x \in X} \{u_n(x) - p^b \cdot x\}.$$

Seller f 's supply correspondence is

$$S_f(p^s) = \arg \max_{y \in Y} \{p^s \cdot y - c_f(y)\},$$

with indirect profit

$$\hat{\pi}_f(p^s) = \max_{y \in Y} \{p^s \cdot y - c_f(y)\}.$$

Interpretation: Ordinary Walrasian equilibrium uses the same price vector for buyers and sellers. The markup equilibrium separates the prices: buyers face $(1 + \alpha)p$, while sellers face p .

2.5 Convex benchmark and Walrasian equilibrium

In convex quasilinear economies, buyer upper contour sets and seller upper contour sets are convex. Equivalently, buyers' valuation functions are concave and sellers' cost functions are convex.

Proposition 1. If all buyers have convex preferences and all sellers have convex technologies, then a feasible allocation x is efficient if and only if there exists $p \in \mathbb{R}_+^L$ such that

$$x_n \in D_n(p) \quad \forall n, \quad y_f \in S_f(p) \quad \forall f,$$

and markets clear in value,

$$\sum_{n \in N} p \cdot x_n = \sum_{f \in F} p \cdot y_f.$$

Interpretation: This is the familiar First Welfare Theorem and Second Welfare Theorem structure for convex quasilinear markets. The rest of the paper asks how much of this can be recovered when convexity fails.

2.6 Measures of nonconvexity

The paper measures nonconvexity by comparing a set S to its convex hull $\text{co}(S)$.

- The **inner radius** of S is

$$r(S) = \sup_{x \in \text{co}(S)} \inf_{T \subseteq S: x \in \text{co}(T)} \text{rad}(T).$$

- The **inner distance** of S is

$$\rho(S) = \sup_{x \in \text{co}(S)} \inf_{y \in S} \|x - y\|.$$

For a convex set, both measures equal zero. The economy-level measures $r_{\mathcal{E}}$ and $\rho_{\mathcal{E}}$ are the largest relevant nonconvexity measures across all buyers' and sellers' upper contour sets.

Interpretation: These measures control how much error is created when an allocation from the convexified economy is rounded back into the original nonconvex feasible sets.

2.7 Shapley–Folkman logic

The Shapley–Folkman Lemma implies that the sum of many nonconvex sets is nearly convex. In this paper, the key implication is that an allocation selected from a convexified economy can be rounded into a true allocation so that only a bounded amount of aggregate imbalance remains.

Interpretation: Individual nonconvexities do not disappear, but in large markets their aggregate effect can be controlled. The markup mechanism uses this logic to turn approximate convexified allocations into feasible nonconvex allocations.

3 Bound-Form First Welfare Theorem

3.1 Rationing loss

If an allocation is not an agent's most preferred choice at a given price, the agent is rationed. The buyer rationing loss is

$$R_n(x, p) = \hat{u}_n(p) - U_n(x, p \cdot x),$$

and the seller rationing loss is

$$R_f(y, p) = \hat{\pi}_f(p) - \pi_f(y, p \cdot y).$$

The total rationing loss is

$$R(x, p) = \sum_{n \in N} R_n(x_n, p) + \sum_{f \in F} R_f(y_f, p).$$

Interpretation: Rationing loss is a money-metric measure of how far an assigned allocation is from being individually optimal at the announced price vector.

3.2 The theorem

Theorem 1 (Bound-Form First Welfare Theorem). For any feasible allocation x and any price vector $p \in \mathbb{R}_+^L$,

$$S^* - S(x) \leq R(x, p) + p \cdot \left(\sum_{f \in F} y_f - \sum_{n \in N} x_n \right).$$

The two terms on the right-hand side are:

- total rationing loss; and
- the value, at prices p , of excess production over consumption.

Interpretation: The theorem gives an efficiency certificate for any allocation-price pair, not just a Walrasian equilibrium. If agents are nearly unrationed and excess production is small, welfare loss is small.

3.3 Relationship to the First Welfare Theorem

At a Walrasian equilibrium:

- buyers and sellers are not rationed, so $R(x, p) = 0$;
- markets clear exactly, so excess production is zero.

Therefore the bound gives

$$S^* - S(x) \leq 0,$$

which implies efficiency.

Interpretation: The ordinary First Welfare Theorem is a special case. The new theorem is a bound-form version: prices certify not exact optimality but a quantitative upper bound on loss.

3.4 Why this theorem is central to the markup mechanism

Markup equilibrium eliminates rationing at the prices agents actually face:

- sellers maximize at price p ;
- buyers maximize at price $(1 + \alpha)p$.

The welfare theorem is applied at price p . Sellers have zero rationing loss at p . Buyers may have positive rationing loss at p because they optimize at $(1 + \alpha)p$, but this loss is small when α is small. Excess production is paid for by the markup and is bounded by the rounding argument.

Interpretation: The theorem converts the engineering features of the mechanism - no seller rationing, small buyer markup, bounded excess supply - into a formal efficiency guarantee.

4 Markup equilibrium and mechanism design

4.1 Definition of markup equilibrium

A markup equilibrium is a triple (x, p, α) where:

- x is a feasible allocation;
- $p \in \mathbb{R}_+^L$ is the producer price vector;
- $\alpha \geq 0$ is a markup parameter.

It satisfies:

1. sellers are paid using p and are not rationed:

$$y_f \in S_f(p) \quad \forall f;$$

2. buyers pay $(1 + \alpha)p$ and are not rationed:

$$x_n \in D_n((1 + \alpha)p) \quad \forall n;$$

3. the budget is weakly balanced:

$$\sum_{n \in N} (1 + \alpha)p \cdot x_n - \sum_{f \in F} p \cdot y_f \geq 0.$$

Interpretation: Sellers and buyers each face linear anonymous prices. The only difference from Walrasian equilibrium is that buyers pay a common proportional markup over the producer price.

4.2 Why the markup works

Without convexity, firms may produce a bundle that cannot be exactly matched by consumer demand at the same price. A markup solves this by reducing buyer demand and creating revenue to cover excess supply.

- If $\alpha = 0$, the markup equilibrium is just a Walrasian equilibrium.
- If Walrasian equilibrium fails to exist, a small positive α can restore feasibility and budget balance.
- The markup is not a lump-sum subsidy; it is financed by buyers' payments.

Interpretation: In electricity or fisheries markets, this is analogous to replacing side payments or uplift payments with a transparent ad valorem markup embedded in market prices.

4.3 Minimal markup mechanism

A minimal markup equilibrium is a markup equilibrium with the smallest feasible markup α .

- If a Walrasian equilibrium exists, the minimal markup is $\alpha = 0$.
- In nonconvex economies, finding the exact minimal markup may be computationally hard.
- The paper therefore develops a simpler mechanism that is computationally tractable and asymptotically efficient.

Interpretation: The minimal markup mechanism is conceptually clean but may not be practical. The simple markup mechanism is designed to be implementable.

4.4 Assumptions for large-market approximation

The paper studies a sequence of economies $\mathcal{E}_t$ and imposes four assumptions:

1. **Limit economy.** The number of buyers grows and the seller-buyer ratio converges to a positive limit.
2. **Bounded nonconvexities.** Individual nonconvexities are uniformly bounded.
3. **Growing gains from trade.** Efficient surplus grows at least linearly with the number of buyers:

$$S_t^* = \Omega(|N_t|).$$

4. **Bounded prices.** Finite choke prices bound relevant demand and supply prices.

Interpretation: These assumptions ensure that individual nonconvexities remain small relative to the scale of the market and that welfare comparisons are meaningful.

4.5 Simple markup mechanism

The simple markup mechanism works as follows.

1. If the reported economy is convex, set $\alpha^* = 0$ and compute a Walrasian equilibrium.
2. Otherwise, fix a candidate $\alpha > 0$.
3. Replace each buyer value u_n by $u_n/(1 + \alpha)$.
4. Replace values by concave envelopes $\text{cav}(u_n)$ and costs by convex envelopes $\text{vex}(c_f)$.
5. Add an operational reserve of size

$$R = \min\{r_{\mathcal{E}}\sqrt{L}, \rho_{\mathcal{E}}L\}.$$

6. Solve the resulting convex program and its dual to obtain approximate allocation $\tilde{x}^\alpha$ and price p^α .
7. Use the Shapley–Folkman Lemma to round $\tilde{x}^\alpha$ to a true nonconvex allocation x^α .
8. Search for the smallest α such that buyer payments cover seller payments:

$$\sum_{n \in N} (1 + \alpha) p^\alpha \cdot x_n^\alpha - \sum_{f \in F} p^\alpha \cdot y_f^\alpha \geq 0.$$

Interpretation: The mechanism replaces a hard nonconvex optimization problem with convexification, rounding, and a one-dimensional search over the markup.

4.6 Approximate efficiency of the simple markup mechanism

Theorem 2. Under Assumptions 1–4:

1. the simple markup mechanism is well-defined;
2. the markup α^* is $O(1/|N_t|)$;
3. the deadweight loss is $O(1)$, so the percentage welfare loss is $O(1/|N_t|)$.

Interpretation: The loss from rounding and excess supply does not grow with market size. Since surplus grows with market size, the percentage loss vanishes.

5 Incentives and computation

5.1 Why incentive compatibility is approximate

In a Walrasian mechanism, an agent can gain from misreporting only if it can move prices. In large markets, one agent has little price impact. The same logic applies to markup mechanisms because agents receive their most preferred bundles at the prices they face.

Interpretation: The key incentive problem is not that agents dislike their allocations; they do not. The only profitable manipulation comes from changing prices or the markup.

5.2 Approximate interim incentive compatibility

The paper studies an independent private values environment with finite type spaces. Theorem 3 states that any markup mechanism is approximately interim incentive compatible:

$$O(|N_t|^{-1/2+\varepsilon})\text{-IIC for any } \varepsilon > 0.$$

Interpretation: Truthful reporting is nearly optimal in expectation before an agent knows all other agents' types. The rate is inherited from strategy-proof-in-the-large results for envy-free mechanisms.

5.3 Approximate ex post incentive compatibility

For ex post incentives, the paper assumes strong monotonicity of expected demand and supply correspondences. Demand is strongly monotone if there exists $m > 0$ such that

$$(d - d') \cdot (p' - p) \geq m \|p - p'\|^2.$$

Supply is strongly monotone if

$$(s - s') \cdot (p - p') \geq m \|p - p'\|^2.$$

Theorem 4. Under strong monotonicity, with probability $1 - O(1/|N_t|)$, the simple and minimal markup mechanisms are

$$O(|N_t|^{-1+\varepsilon})\text{-EPIC for any } \varepsilon > 0.$$

Interpretation: Strong monotonicity makes aggregate demand and supply responsive to prices. This limits the price impact of any single participant and yields a stronger incentive bound.

5.4 Computational advantages

- Exact efficient allocation in nonconvex markets can be computationally hard.
- Conditional on a markup α , the simple markup mechanism solves a convex optimization problem.
- The markup parameter can be searched over with a one-dimensional line or binary search.
- The mechanism requires little more communication than ordinary Walrasian mechanisms: allocation, producer prices, and one markup parameter.

Interpretation: The mechanism is designed to be practical in markets like electricity, fisheries rights, or spectrum-like settings where fixed costs or indivisibilities make full nonconvex optimization difficult.

6 Applications and interpretation

6.1 Fixed costs and start-up costs

The paper's one-good example features firms that can produce zero or produce up to capacity only after paying a fixed cost. Such fixed costs create discontinuous supply, so ordinary supply and demand may not intersect.

Interpretation: This is the canonical market-design problem: marginal prices do not cover fixed costs, and ordinary competitive equilibrium can fail.

6.2 Electricity markets

Electricity markets often have nonconvexities from start-up costs, minimum generation levels, ramp constraints, and locational constraints. Existing mechanisms frequently use uplift payments to make generators whole.

Interpretation: The markup mechanism avoids budget deficits from uplift by charging buyers a markup over producer prices. This reduces incentives for producers to exaggerate fixed costs to receive uplift.

6.3 Fisheries and other capacity-reduction markets

The paper discusses motivation from fisheries rights mechanisms where achieving a sustainable catch requires some boats to exit, creating fixed-cost losses.

Interpretation: Markups can help implement feasible market-based reallocation while preserving anonymous linear prices and avoiding personalized transfers.

6.4 Relation to approximate equilibrium literature

Classical Shapley–Folkman approximations often yield allocations that are only approximately feasible or may ration agents. The markup mechanism instead insists on actual feasibility, no rationing at the relevant prices, and no budget deficit.

Interpretation: The paper shifts the approximation target from approximate feasibility to approximate efficiency and approximate incentive compatibility.

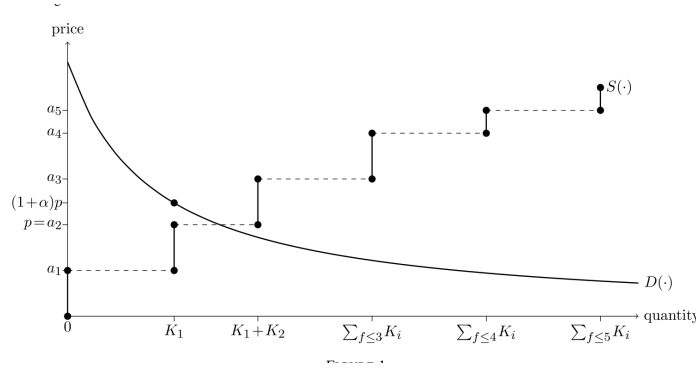
7 Tables, figures, and original equation panels

7.1 Figure 1: Single-product example

Read:

- The upward step function is aggregate supply from firms with fixed costs.
- The downward curve is continuous demand.
- Ordinary Walrasian equilibrium may not exist because the demand curve need not intersect the discontinuous supply correspondence.
- A markup equilibrium can choose a producer price $p = a_2$, assign production K_1 , and set the buyer markup so that demand at $(1 + \alpha)p$ equals K_1 .

Economic message: The figure shows why the markup is needed: it creates a wedge between buyer and seller prices that restores feasibility and budget balance.

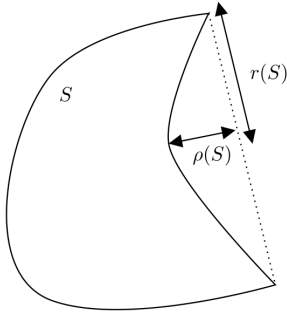


7.2 Figure 2 and model primitives

Read:

- Figure 2 illustrates the inner radius $r(S)$ and inner distance $\rho(S)$ of a nonconvex set.
- The original model panel defines type distributions, feasibility, and surplus.
- These objects determine the size of the operational reserve used in the simple markup mechanism.

Economic message: Nonconvexity measures quantify how much error is introduced when convexified allocations are rounded into feasible nonconvex allocations.



(a) Figure 2: nonconvexity measures.

of $\mathcal{E}$ of buyers and sellers are private information, not directly observable by the designer. The designer knows $|N|$ and $|F|$, as well as a space of possible valuation functions for the buyers, $\mathcal{U}$, and a space of cost functions for the sellers, $\mathcal{C}$, both assumed to be additive of [Aumann \(1963\)](#).⁹ Let the normalized counting measures μ on $\mathcal{U}$ and χ on $\mathcal{C}$ be defined by

$$\mu(u_n) := \frac{\# \text{ of buyers in } \mathcal{E} \text{ with valuation function } u_n}{|N|}$$

$$\chi(c_f) := \frac{\# \text{ of sellers in } \mathcal{E} \text{ with cost function } c_f}{|F|},$$

and let $\phi := |F|/|N|$, so that $\langle N, \mu, \phi, \chi \rangle$ is an alternative specification of economy $\mathcal{E}$.

Allocations and efficiency. An allocation $\mathbf{x} = ((x_n)_{n \in N}, (y_f)_{f \in F})$ is an assignment of bundles $x_n \in X$ to each buyer $n \in N$ and production bundles $y_f \in Y$ for each seller $f \in F$. An allocation is (resource-)feasible if it satisfies $\sum_{n \in N} x_n \leq \sum_{f \in F} y_f$. The surplus $\mathcal{S}(\mathbf{x})$ of allocation $\mathbf{x} \in \mathbf{X}$ is

$$\mathcal{S}(\mathbf{x}) := \sum u_n(x_n) - \sum c_f(y_f).$$

(b) Model primitives and surplus.

7.3 Pricing, convexity, and Shapley–Folkman

Read:

- The pricing panel defines buyer demand, seller supply, indirect utility, and indirect profit.
- Convex quasilinear economies admit Walrasian equilibrium and the usual welfare results.
- The Shapley–Folkman panel shows the near-convexity result used for rounding convexified allocations.

Economic message: The mechanism borrows two tools: Walrasian pricing from convex economies and Shapley–Folkman rounding from nonconvex economies.

2.1.2. Pricing rules. To prepare for our markup equilibrium, we allow two different price vectors $p^b, p^s \in \mathbb{R}_+^L$ for buyers and sellers, such that a buyer n pays $p^b \cdot x$ for bundle x , and a seller f receives $p^s \cdot y$ for supplying y .

Denote buyer n 's demand correspondence by $D_n : \mathbb{R}_+^L \rightrightarrows X$, mapping a price vector p^b to the set of utility-maximizing bundles $D_n(p^b)$, which is well-defined because u_n is upper semicontinuous. The indirect utility at p^b is $\bar{u}_n(p^b) := \max_{x \in X} \{u_n(x) - p^b \cdot x\}$. Similarly, denote seller f 's supply correspondence by $S_f : \mathbb{R}_+^L \rightrightarrows Y$, mapping a price vector p^s to the set of profit-maximizing bundles $S_f(p^s)$, which is well-defined because c_f is lower semicontinuous. The indirect profit at p^s is $\bar{\pi}_f(p^s) := \max_{y \in Y} \{p^s \cdot y - c_f(y)\}$.

2.2. Convex quasilinear economies

We study convexity as it pertains to the set of payoff-improving allocations for an agent in the economy. The $\bar{u}$ -upper contour set of buyer $n \in N$ is

$$UC_n^{\bar{u}} := \{(x, t) \in X \times \mathbb{R} : U_n(x, t) \geq \bar{u}\},$$

while the $\bar{\pi}$ -upper contour set of seller $f \in F$ is

$$UC_f^{\bar{\pi}} := \{(y, t) \in Y \times \mathbb{R} : \pi_f(y, t) \geq \bar{\pi}\}.$$

Buyer n has convex preferences if X is convex and the upper contour set $UC_n^{\bar{u}}$ is convex for all $n \in N$. Similarly, seller f has convex preferences if Y is convex and the upper contour set $UC_f^{\bar{\pi}}$ is convex for all $f \in F$.

(a) Pricing rules and convex upper contour sets.

⁹When agents' upper contour sets are not convex, competitive equilibrium might not exist. Proposition 2, the Shapley–Folkman Lemma, helps identify allocations that are nearly competitive equilibria.¹¹

Proposition 2 (Shapley–Folkman Lemma). Let $S_i \subseteq \mathbb{R}^L$ for $i = 1, \dots, M$, and let $S = \bigoplus_{i=1}^M S_i$ be the Minkowski sum of those sets. Then any $x \in \text{co}(S)$ may be written as $x = \sum_{i=1}^M x_i$ where $x_i \in \text{co}(S_i)$ and $\|x_i - \text{co}(S_i)\| \leq L' := \min(L, M)$. Moreover, there exists $y, y' \in S$ such that $\|x - y\| \leq (\max_i r(S_i))\sqrt{L}$ and $\|x - y'\| \leq (\max_i \rho(S_i))L'$.

Proposition 2 has been used to establish results about approximate equilibria, which are

(b) Shapley–Folkman lemma.

7.4 Bound-Form First Welfare Theorem

Read:

- The panel defines buyer and seller rationing losses.
- The theorem bounds deadweight loss by rationing loss plus the value of budget deficit/excess production.
- For a Walrasian equilibrium, both terms are zero, recovering efficiency.

Economic message: This theorem is the paper's main analytical tool: it turns no-rationing and bounded excess supply into a welfare guarantee.

bundle at the given prices over the payoff it receives in its assigned allocation.

Definition 1 (Rationing loss). The *rationing loss* $\mathcal{R}_n(x, p)$ of buyer n at allocation x given price p is

$$\mathcal{R}_n(x, p) := \hat{u}_n(p) - U_n(x, p \cdot x).$$

The *rationing loss* $\mathcal{R}_f(y, p)$ of seller f at allocation y given price p is

$$\mathcal{R}_f(y, p) := \hat{\pi}_f(p) - \pi_f(y, p \cdot y).$$

The *rationing loss* of allocation $\mathbf{x} = ((x_n)_{n \in N}, (y_f)_{f \in F})$ given price p is

$$\mathcal{R}(\mathbf{x}, p) := \sum_{n \in N} \mathcal{R}_n(x_n, p) + \sum_{f \in F} \mathcal{R}_f(y_f, p).$$

If competitive equilibrium does not exist, any allocation-price pair must entail rationing, wasted supply (leading to a budget deficit), or both. Our first main result establishes that the magnitude of such rationing and budget losses fully characterizes the efficiency of the allocation.

Theorem 1 (Bound-Form First Welfare Theorem). Let $p \in \mathbb{R}_+^L$ be a price vector and $\mathbf{x} \in \mathbf{X}$ be any feasible allocation. Then, the deadweight loss of $\mathbf{x}$ is bounded by

$$\underbrace{S^* - S(\mathbf{x})}_{\text{deadweight loss}} \leq \underbrace{\mathcal{R}(\mathbf{x}, p)}_{\text{rationing loss}} + p \cdot \left(\sum_{f \in F} y_f - \sum_{n \in N} x_n \right).$$

7.5 Markup equilibrium and simple markup mechanism

Read:

- Markup equilibrium requires sellers to maximize at p , buyers to maximize at $(1 + \alpha)p$, and the budget to be weakly balanced.
- The simple markup mechanism constructs a convexified economy, adds an operational reserve, rounds by Shapley–Folkman, and searches for the smallest budget-feasible markup.
- Theorem 2 gives asymptotic efficiency: $\alpha^* = O(1/|N_t|)$ and percentage welfare loss is $O(1/|N_t|)$.

Economic message: Markup equilibrium is an implementable approximation to Walrasian equilibrium for nonconvex markets.

Definition 2 (Markup equilibrium). A *markup equilibrium* is a triple $(\mathbf{x}, p, \alpha)$ consisting of a resource-feasible allocation $\mathbf{x} \in \mathbf{X}$, a price $p \in \mathbb{R}_+^L$, and a markup parameter $\alpha \geq 0$ such that:

- (a) sellers' payments are determined by price vector p , and sellers are not rationed at these prices, so $y_f \in S_f(p)$;
- (b) buyers' payments are determined by price vector $(1 + \alpha)p$, and buyers are not rationed at these prices, so $x_n \in D_n((1 + \alpha)p)$; and
- (c) budgets are at least weakly balanced, so $\sum_{n \in N} (1 + \alpha)p \cdot x_n - \sum_{f \in F} p \cdot y_f \geq 0$.

A *markup mechanism* is a mechanism that inputs reports of cost and value functions and outputs a markup equilibrium. With finite choke prices on supply and demand, markup equilibria with no trade and $\alpha = \infty$ always exist. However, we focus on markup mechanisms that select α close to zero and result in few unallocated goods, because their allocations are nearly efficient. This follows from Theorem 1 applied to allocation-price pair $(\mathbf{x}, p)$: if few goods are unallocated, the budget deficit at price p is small, while if prices p and $(1 + \alpha)p$ are close, the

(a) Markup equilibrium definition.

Definition 3 (Simple markup mechanism). The *simple markup mechanism* is the markup mechanism with parameters $(\mathbf{x}^*, p^*, \alpha^*)$ determined as follows. If all reported values are concave and all reported costs are convex, set $\alpha^* = 0$ and choose $(\mathbf{x}^*, p^*)$ to be some Walrasian equilibrium. Otherwise, for each $\alpha > 0$, consider the following convex program:

$$\min_{p \in \mathbb{R}_+^L, x_n \in \text{co}(\mathbf{X}), y_f \in \text{co}(\mathbf{Y})} \sum_{n \in N} \frac{\text{cav}(u_n)(x_n)}{1 + \alpha} - \sum_{f \in F} \text{vex}(c_f)(y_f) - p \cdot \left(\sum_{n \in N} x_n + R 1_L - \sum_{f \in F} y_f \right),$$

where 1_L is the vector of ones in $\mathbb{R}^L$. Let $(\bar{\mathbf{x}}^\alpha, \bar{p}^\alpha)$ denote any solution to this program.

From $\bar{\mathbf{x}}^\alpha \in \text{co}(\mathbf{X})$, obtain, via Proposition 2, an allocation $\mathbf{x}^\alpha \in \mathbf{X}$ with $\|\mathbf{x}^\alpha - \bar{\mathbf{x}}^\alpha\| \leq R$ such that $x_n^\alpha \in D_n((1 + \alpha)p)$ for each $n \in N$ and $y_f^\alpha \in S_f(p)$ for each $f \in F$. Let

$$\alpha^* := \min \left\{ \alpha \left| \sum_{n \in N} (1 + \alpha)p^\alpha \cdot x_n^\alpha - \sum_{f \in F} p^\alpha \cdot y_f^\alpha \geq 0 \right. \right\}, \quad (\text{A})$$

and set $p^* = p^{\alpha^*}$ and $\mathbf{x}^* = \mathbf{x}^{\alpha^*}$.

Theorem 2 formalizes the preceding informal argument, showing that the simple markup mechanism is well-defined and $O(1/|N_t|)$ -efficient.

Theorem 2 (Approximate efficiency of simple markup mechanism). Let $\mathcal{E}_t$ be a sequence of economies satisfying Assumptions 1–4. Then:

- (a) the simple markup mechanism is well-defined (that is, the minimum in (A) is attained),
- (b) the simple markup mechanism's markup α^* is $O(1/|N_t|)$, and

(b) Simple markup mechanism and Theorem 2.

7.6 Incentive results

Read:

- Theorem 3 establishes approximate interim incentive compatibility for markup mechanisms in finite-type IPV economies.
- Theorem 4 strengthens the incentive result under strong monotonicity of expected demand and supply.
- The economic logic is that one agent has little effect on prices in a large market.

Economic message: The mechanism is not exactly strategy-proof in finite markets, but the possible gain from manipulation vanishes as the market grows.

mechanism. A mechanism is ε -IIC if, under truthful reporting, each agent's expected payoff from any report is no more than ε greater than the expected payoff of the truthful report.

As markup mechanisms do not ration producers or consumers at their respective prices, all markup mechanisms are envy-free. As a result, Theorem 1 of Azevedo and Budish (2019) implies that every markup mechanism is strategy-proof in the large, leading to the following asymptotic interim incentive compatibility result in the IPV model with finite type spaces.

Theorem 3 (Approximate IIC). In an IPV economy with finite type spaces $\mathcal{U}$ and $\mathcal{C}$, any markup mechanism is $O(1/|N_t|^{1/2-\varepsilon})$ -IIC for any $\varepsilon > 0$.

(a) Theorem 3: approximate IIC.

its price vector. As Watt (2025) shows for Walrasian mechanisms, an agent's ability to influence prices with any report is approximately inversely proportional to market size when the expected demand and supply correspondences are *strongly monotone*, as defined below.

Definition 4 (Strong monotonicity). The expected demand correspondence D_μ is *strongly monotone* if there exists some $m > 0$ such that for all $p, p' \in \mathcal{P}$, $d \in D_\mu(p)$, and $d' \in D_\mu(p')$,

$$(d - d') \cdot (p' - p) \geq m \|p - p'\|^2.$$

Similarly, the expected supply correspondence S_χ is *strongly monotone* if there exists some $m > 0$ such that for all $p, p' \in \mathcal{P}$, $s \in S_\chi(p)$, and $s' \in S_\chi(p')$,

$$(s - s') \cdot (p - p') \geq m \|p - p'\|^2.$$

Under the assumption of strong monotonicity, the expected demand and supply correspondences are responsive to small price changes, and we have the following ex post incentive properties of the simple and minimal markup mechanisms.

Theorem 4 (Approximate EPIC). In an IPV economy with strongly monotone expected demand and supply correspondences satisfying Assumption 3 almost surely, with probability $1 - O(1/|N_t|)$ over draws of $\mathcal{E}_t$, the simple and minimal markup mechanisms are $O(1/|N_t|^{1-\varepsilon})$ -EPIC for any $\varepsilon > 0$.

(b) Definition 4 and Theorem 4: approximate EPIC.

8 Short summary

- Ordinary Walrasian equilibrium may fail in nonconvex markets.
- Markup equilibrium restores feasibility by allowing buyers to pay a proportional markup over producer prices.
- Sellers face price p , buyers face price $(1 + \alpha)p$.

- The mechanism is resource-feasible, budget-feasible, individually rational, and envy-free at the relevant prices.
- The Bound-Form First Welfare Theorem bounds welfare loss by rationing loss plus the value of excess production.
- The simple markup mechanism uses convexification, an operational reserve, Shapley–Folkman rounding, and a one-dimensional search for α .
- Under large-market assumptions, the markup is $O(1/|N|)$ and percentage efficiency loss is $O(1/|N|)$.
- Incentive compatibility is approximate because a single participant has little effect on prices in a large market.
- The framework is especially relevant for markets with fixed costs, indivisibilities, start-up costs, or other nonconvex production technologies.

9 Conclusion

9.1 Core conclusion

Milgrom and Watt show that anonymous linear price mechanisms can be extended to nonconvex markets by adding a simple markup. The markup equilibrium keeps the appealing properties of Walrasian equilibrium - linear prices, individual optimization, feasibility, and budget balance - while tolerating fixed costs and other nonconvexities.

9.2 Economic intuition

Nonconvexities create gaps between supply and demand at any common price. Instead of subsidizing firms through uplift payments or assigning agents bundles they do not demand, the mechanism lets buyers pay slightly more than sellers receive. This extra revenue covers excess production created by rounding and nonconvexity. In large markets, the required markup is small.

9.3 Takeaway

The paper reframes nonconvex market design. The goal is not to force exact Walrasian equilibrium where it cannot exist, but to preserve the informational and incentive advantages of prices while achieving feasible, budget-balanced, nearly efficient outcomes.